



Barker
College

2021

YEAR 12
ASSESSMENT BLOCK

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Student Number

Mathematics Advanced

Staff Involved:

• AYG • LZM • LAK • DXC • PJR
• RJW* • PDJ • DZP • ARP

FRIDAY 26th MARCH, NOON

140 copies

TOPICS COVERED:

Year 11 Content	Sequences & Series
Series & Finance	Graphs & Equations
Curve Sketching using the Derivative	Integration

General

Instructions:

- Working time - 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A separate reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and / or calculations
- Marks may not be awarded for careless or poorly arranged working

Total marks:

80

Section I - 8 marks (pages 2 - 4)

- Attempt Multiple Choice Questions 1 - 8
- Allow about 15 minutes for this section

Section II - 72 marks (pages 5 - 22)

- Attempt Questions 9 - 17
- Show all necessary working
- Allow about 1 hours and 45 minutes for this section

Section I

8 marks

Attempt Questions 1 – 8

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 8.

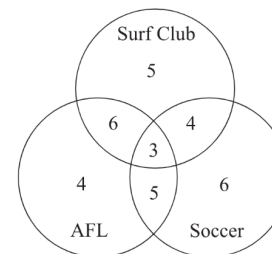
1 Using the log laws, fully simplify: $9 \log_a(\sqrt[3]{a})$

- A. 3
- B. 27
- C. $\sqrt[3]{9}$
- D. $3 \log_a(a)$

2 The sum of the first 10 terms of the series: $100 - 80 + 64 - \dots$ is closest to:

- A. -55
- B. 50
- C. 55
- D. 446

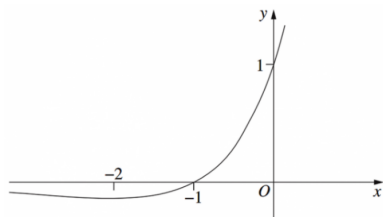
3 In a classroom, students are asked what sports club they are members of and the results are shown in the Venn diagram.



A student who is a member of a soccer club is chosen at random. What is the probability that he/she is also a member of a surf club?

- A. $\frac{2}{5}$
- B. $\frac{4}{11}$
- C. $\frac{2}{9}$
- D. $\frac{7}{18}$

- 4 The diagram shows the graph of $y = e^x(1+x)$



How many solutions are there to the equation $e^x(1+x) = 1-x^2$?

- A. 0
 B. 1
 C. 2
 D. 3
- 5 A circle with centre $(a, -2)$ and radius 5 units has equation $x^2 - 6x + y^2 + 4y = b$ where a and b are real constants.

The values of a and b are respectively:

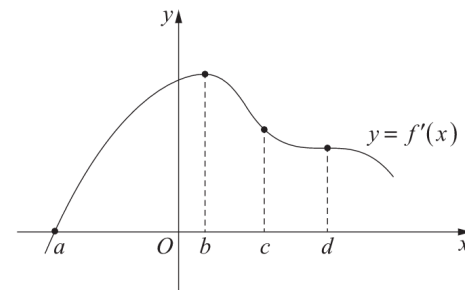
- A. -3 and 38
 B. 3 and 12
 C. -3 and -8
 D. 3 and 18

- 6 Let $f(x) = x^2$

Which one of the following is **not** true

- A. $f(ab) = f(a)f(b)$
 B. $f(a) - f(-a) = 0$
 C. $f(2a) = 4f(a)$
 D. $f(a-b) = f(a) - f(b)$

- 7 The diagram shows the graph of $f'(x)$, the derivative of a function.



For what value of x does the graph of the function $f(x)$ have a point of inflexion?

- A. $x = a$
 B. $x = b$
 C. $x = c$
 D. $x = d$
- 8 The function $y = f(x)$ has a turning point at $(4, 2)$. It is transformed to $y = -2f(2x+6) + 3$. Where is the new turning point?
- A. $(-1, -1)$
 B. $(-5, -2)$
 C. $(-5, -1)$
 D. $(1, -1)$

End of Section I

Student Number

Allow about 1 hour and 45 minutes for this section

In Questions 9 – 17, your responses should include relevant mathematical reasoning and/or calculations.

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Student Number

Question 10 (8 marks)

On any given day, the number X of telephone calls that Luke receives is a random variable with probability distribution given in the table below.

2

- (i) Show that $p = 0.1$ and hence complete the discrete probability distribution table, assuming no other values of x are possible.

x	0	1	2	3	4	5
$P(X = x)$	$p =$	0.15	$2p =$	$3p =$	0.2	0.05

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- (ii) Find $E(X)$.

2

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- (iii) What is the probability that Luke receives more than the mean number of phone calls on a given day?

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- (iv) Calculate the $Var(X)$.

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- (v) Calculate the standard deviation, to 2 decimal places.

1

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Student Number

Question 11 (8 marks)

- (a)
- (i) Sketch the curve $f(x) = (x - 1)(x + 1)(x + 3)$, showing x and y intercepts, in the space below. No need to find turning points. 2

- (ii) Hence or otherwise, solve $f(x) \leq 0$, giving your answer in interval notation. 2
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Question 11 (continued)

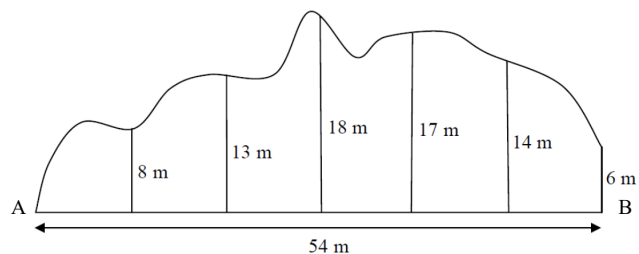
- (b) Evaluate the following given that $f(x) = \sqrt{1 - x}$ and $g(x) = \frac{1}{x}$
- (i) $g(f(-3))$ 1
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-
-
-
- (ii) Determine the domain and range of $g(f(x))$ 3
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Question 12 (8 marks)

Student Number

- (a) The sketch shows a piece of land which borders the side of a straight road AB. 3

The length of AB is 54m. At equal intervals along AB, perpendicular measurements are made to the boundary as shown in the sketch.



Use the trapezoidal rule with 7 function values to find the approximate area of the piece of land.

[illegible]

Question 12 (continued)

- (b) (i) Prove the identity $(1 - \sin \theta)(1 + \sin \theta) = \cos^2 \theta$ 2

[illegible]

- (ii) Hence solve the equation $(1 - \sin \theta)(1 + \sin \theta) = 0.5$ for $0^\circ \leq \theta \leq 360^\circ$ **3**

[illegible]

Student Number

3

2

2

1

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Student Number

Question 14 (8 marks)

(a) State the transformations that have occurred to the following graphs from the originals. The order may or may not be important.

(i) From $y = \ln x$ to $y = -\ln(x + 3)$ 1

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(ii) From $y = e^x$ to $y = 2e^x - 5$ 1

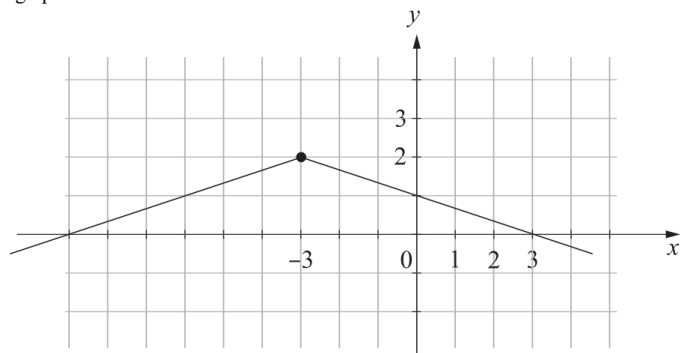
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(b) The function $f(x) = |x|$ is transformed and the equation of the new function is $y = kf(x + b) + c$.

The graph of the new function is shown below.



Determine the values of k , b and c . 2

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Question 14 (continued)

(c) (i) Use differentiation by first principles to find $f'(x)$ given $f(x) = 2x^2 + 5x$. 2

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

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(ii) Hence, find the equation of the tangent to the curve $y = 2x^2 + 5x$ at the point on the curve where $x = 1$. 2

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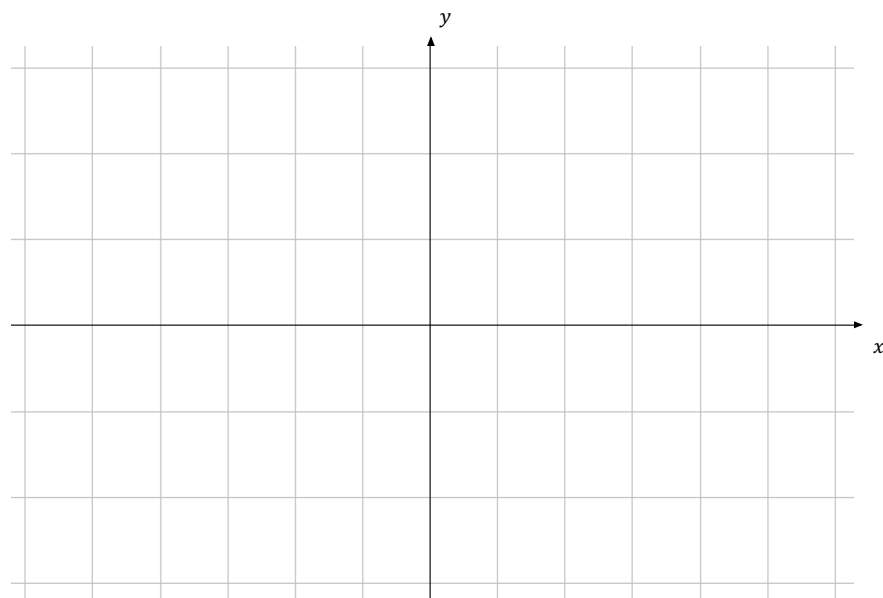
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Student Number

Question 15 (8 marks)

- (a) Sketch the curve $y = 2\cos x$ in the interval $[-2\pi, 2\pi]$. Label x and y intercepts. 2



Question 15 (continued)

- (b) Lucy borrows \$500 000 to be repaid at a reducible interest rate of 0.6% per month. Let A_n be the amount owing at the end of n months and M be the monthly repayment.

- (i) Show that $A_2 = 500000(1.006)^2 - M(1 + 1.006)$ 1

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- (ii) Show that $A_n = 500000(1.006)^n - M\left(\frac{1.006^n - 1}{0.006}\right)$ 2

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- (iii) Lucy budgets a monthly repayment of \$3500. Determine how many months it would take her to pay off the loan. 3

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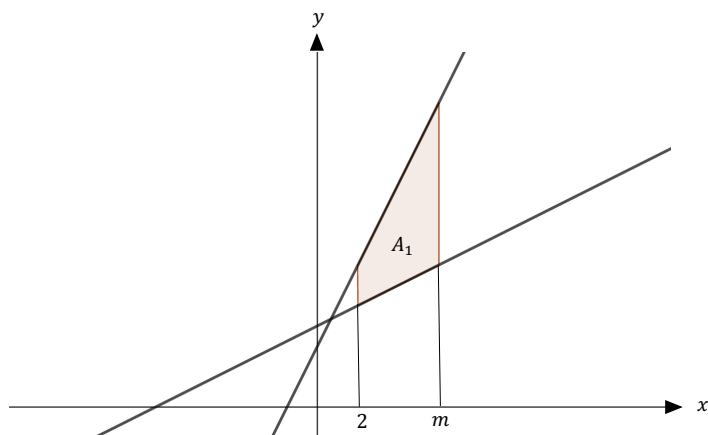
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Student Number

(a)

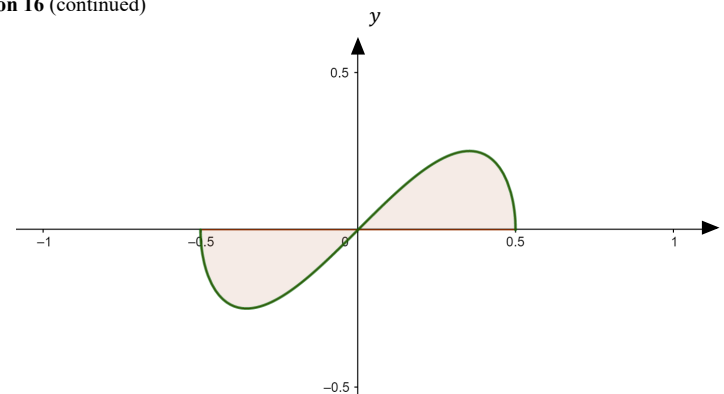


Given that $\text{area } A_1 = 20 \text{ units}^2$, find the value of m marked on the diagram.

3

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(b)



(i) Find $\frac{d}{dx}(1 - 4x^2)^{\frac{3}{2}}$ and express your answer in surd form.

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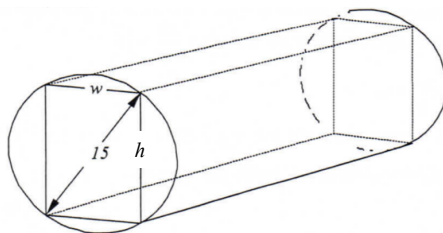
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[illegible]

Question 17 (8 marks)

Student Number



(i) Show that $S = k(225w - w^3)$

1

(ii) Hence, what dimensions will give a beam of maximum strength? Justify your answer. 4

4

Question 17 (continued)

- (iii) A square beam with a diagonal of 15 cm could have been cut from the log.

Show that the rectangular beam of maximum strength is more than 8% stronger than this square beam.

3

End of Paper

year 12 Advanced Ass #2 Solutions

26/03/21

Section 1

① $9 \log(a^{1/3})$
 $= \frac{1}{3} (9 \log a)$
 $= 3(1)$
 $= 3$

(A)

② $S_{10} = 100(1 - (-\frac{4}{5})^{10})$
 $1 + \frac{4}{5}$
 $= 49.59 \dots$

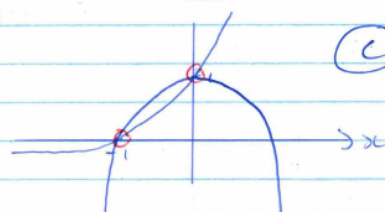
(B)

③ $7/18$

(D)

④

(C)



⑤ $x^2 + 6x + 9 + y^2 + 4y + 4 = 13 + b$
 $(x+3)^2 + (y+2)^2 = 13 + b$
 Centre $(-3, -2)$ $R = \sqrt{13+b}$
 $25 - 13 = b$
 $12 = b$

(B)

(D)

(B)

⑥ $y = -2f(2(x+3)) + 3$
 $(4, 2)$
 $x + \frac{1}{2} \rightarrow (4, -4) \times 2$
 $(2, -4)$
 $x - 3 \rightarrow (-1, -4)$
 $(-1, -1) \uparrow +3$

(A)

Section 2

④ (a) $\frac{dy}{dx} = 6(3x^2 - 4)^5 \times 6x$
 $= 36x(3x^2 - 4)^5$

(B) $u = x \quad v = e^x$
 $u' = 1 \quad v' = e^x$
 $\frac{dy}{dx} = xe^x + e^x$
 $= e^x(x+1)$

(Y) $u = (x-1)^4 \quad v = x^2$
 $u' = 4(x-1)^3 \quad v' = 2x$
 $\frac{dy}{dx} = \frac{4x^2(x-1)^3 - 2x(x-1)^4}{x^4}$
 $= \frac{2x(x-1)^3(2x - (x-1))}{x^4}$
 $= \frac{2(x-1)^3(x+1)}{x^3}$

9(b)(2) $3x^2 - \frac{x^4}{2} + C$

(B) $\frac{(3x-2)^{-3}}{-3 \times 3} + C$
 $= -\frac{(3x-2)^{-3}}{9} + C$

10(i) $p + 0.15 + 2p + 3p + 0.2 + 0.05 = 1$
 $6p = 0.6$
 $p = 0.1$

$\therefore 2p = 0.2 \quad 3p = 0.3$

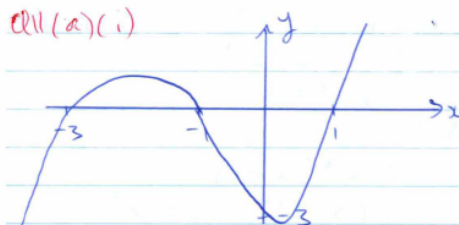
(ii) $E(X) = 2.5$

(iii) $P(X > 2.5) = 0.3 + 0.2 + 0.05 = 0.55$

(iv) $\text{Var}(X) = 8 - 1 - (2.5)^2 = 1.85$

(v) $\sigma = \sqrt{1.85} = 1.36$

11(a)(i)



(ii) $(-\infty, -3] \cup [-1, 1]$

(b)(i) $f(-3) = \sqrt{1+3} = 2$

$g(2) = \frac{1}{2}$

(ii) $g(\sqrt{1-x}) = \frac{1}{\sqrt{1-x}}$

D: $x < 1$

R: $y > 0$

12(a) $A \approx \frac{b-a}{2n} = \frac{54-0}{2(6)} = 4.5$

$A \approx 4.5(0 + 2(8+13+18+17+14) + 6)$
 $\approx 4.5(146)$
 $\approx 657 \text{ m}^2$

(b)(i) $1 - \sin^2 \theta = \cos^2 \theta$
 From Ref Sheet $\sin^2 \theta + \cos^2 \theta = 1$
 $\cos^2 \theta = 1 - \sin^2 \theta$

$\therefore \cos^2 \theta = \cos^2 \theta$
 $\cos \theta = \pm \frac{1}{\sqrt{2}}$
 $\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$

13(i) $y' = 3x^2 - 3x - 6$
 $x^2 - x - 2 = 0$
 $(x-2)(x+1) = 0$
 $x = 2, -1$

$x = 2 \quad y = -5 \quad | \quad x = -1 \quad y = 8.5$
 $(2, -5) \quad | \quad (-1, 8.5)$

$y'' = 6x - 3$
 $x = 2$
 $y'' = 6(2) - 3 = 9 > 0$ C. up
 $(2, -5)$ L. Min

$x = -1$
 $y'' = 6(-1) - 3 = -9 < 0$ C. Down
 $(-1, 8.5)$ L. Max

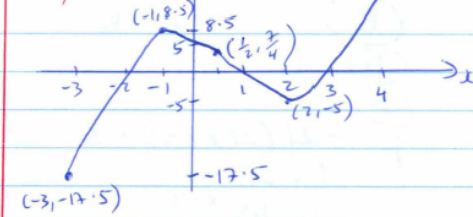
(ii) $6x - 3 = 0$
 $x = \frac{1}{2}$

$y = 7/4$

x	0	1/2	1
y''	-3	0	3

 $< 0 \quad > 0$

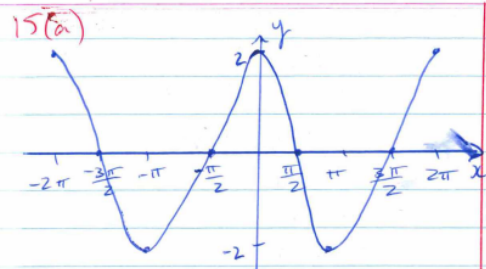
$\therefore$ Change in concavity
 $(1/2, 7/4)$ P.O.T



(iv) $y = 21$

14(a)(i) Reflect x-axis, left 3
 (ii) Vertical dilation by 2, down 5 spaces

(b) $k = -1/3$
 $b = 3$
 $c = 2$



(b) (i) $A_1 = 500000(1.006) - m$
 $A_2 = (500000(1.006) - m)(1.006) - m$
 $= 500000(1.006)^2 - m(1.006) - m$
 $= 500000(1.006)^2 - m(1 + 1.006)$

(ii) $A_N = 500000(1.006)^N - m(1 + 1.006 + \dots + 1.006^{N-1})$
 $= 500000(1.006)^N - m \left(\frac{1(1.006^N - 1)}{1.006 - 1} \right)$

$A_N = 500000(1.006)^N - m \left(\frac{1.006^N - 1}{0.006} \right)$

(iii) $0 = 500000(1.006)^N - 3500 \left(\frac{1.006^N - 1}{0.006} \right)$
 $3500 \left(\frac{1.006^N - 1}{0.006} \right) = 500000(1.006)^N$

$3500(1.006^N) - 3500 = 3000(1.006)^N$

$3500(1.006)^N - 3000(1.006)^N = 3500$

$500(1.006)^N = 3500$

$(1.006)^N = 7$

$N = \ln(7)$

$\ln(1.006)$

$N = 325.29 \dots$

$N = 326$

16(a) $\int_2^m (2x + 3 - \frac{1}{2}x - 4) dx = 20$

$= \int_2^m (\frac{3}{2}x - 1) dx = 20$

$= \left[\frac{3x^2}{4} - x \right]_2^m = 20$

$\frac{3m^2}{4} - m - \left(\frac{3(2)^2}{4} - 2 \right) = 20$

$\frac{3m^2}{4} - m - 1 = 20$

$3m^2 - 4m - 84 = 0$

$(3m + 14)(m - 6) = 0$

$m = \frac{-14}{3}$ $m = 6$

(b) (i) $\frac{dy}{dx} = \frac{3}{2}(1 - 4x^2)^{1/2}(-8x)$

$= -12x\sqrt{1 - 4x^2}$

(ii) $2 \left(\frac{-1}{12} \right) \int_0^{0.5} -12x\sqrt{1 - 4x^2} dx$

$= -\frac{1}{6} \int_0^{0.5} (1 - 4x^2)^{3/2} dx$

$= -\frac{1}{6} (1 - 4(0.5)^2)^{3/2} - (1 - 4(0)^2)^{3/2}$

$= -\frac{1}{6} (0 - 1)$

$= \frac{1}{6}$

OR

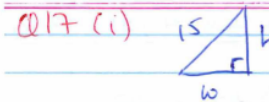
$2 \int_0^{0.5} x\sqrt{1 - 4x^2} dx$

$= 2 \left[\frac{x(1 - 4x^2)^{3/2}}{(-8x)(3/2)} \right]_0^{0.5}$

$= \frac{2}{-12} \left[(1 - 4(0.5)^2)^{3/2} - (1 - 4(0)^2)^{3/2} \right]$

$= \frac{2}{-12} (0 - 1)$

$= \frac{1}{6}$



$225 = w^2 + h^2$

$225 - w^2 = h^2$

$S = k(225 - w^2)(w)$

$= k(225w - w^3)$

(ii) $S = 225kw - kw^3$

$S' = 225k - 3kw^2$

$225k - 3kw^2 = 0$

$3kw^2 = 225k$

$w = \pm 5\sqrt{3}$

$S'' = -6kw$

$w = 5\sqrt{3}$

$= -6k(5\sqrt{3})$

$= -30k\sqrt{3}$

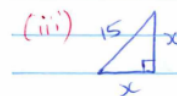
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$w = 5\sqrt{3}$ is L. Max

$225 - (5\sqrt{3})^2 = d^2$

$150 = d^2$

$d = 5\sqrt{6}$



$2x^2 = 225$

$x^2 = \frac{225}{2}$

$x = \frac{15\sqrt{2}}{2}$

Rectangle: $S = k(5\sqrt{6})^2(5\sqrt{3})$

$= 750\sqrt{3}k$

Square:

$S = k \left(\frac{15\sqrt{2}}{2} \right)^2 \left(\frac{15\sqrt{2}}{2} \right)$

$= \frac{3375\sqrt{2}k}{4}$

$= \frac{750\sqrt{3}k}{4}$

$= \frac{3375\sqrt{2}k}{4}$

$= 750\sqrt{3}k \times \frac{4}{3375\sqrt{2}k}$

$= 1.08866 \dots$

$\therefore 8.866\% \text{ more}$